

E-Commerce Conversion Prediction

Model Development & Validation Summary

Objective

Predict whether a user will convert (purchase) based on demographic, browsing, and campaign-related features, using a labeled training set to build a model that generalizes to unseen (private) test users.

Data Preparation

- Missing numeric values imputed with the median (plus a missing-value indicator); missing categorical values imputed with the most frequent category.
- Categorical features (Device_Type, Traffic_Source) encoded with One-Hot Encoding; numeric features standardized (z-score).
- Feature engineering: Engagement_Rate (Products_Viewed / Pages_Viewed), Time_Per_Page (Time_On_Site / Pages_Viewed), and Log_Income (log1p of Income) added to help the linear model capture non-linear and skewed relationships.
- Target is imbalanced (~69% / 31% non-converters vs. converters), so F1-score — not accuracy — was used as the primary evaluation metric, and class weighting was applied where supported.

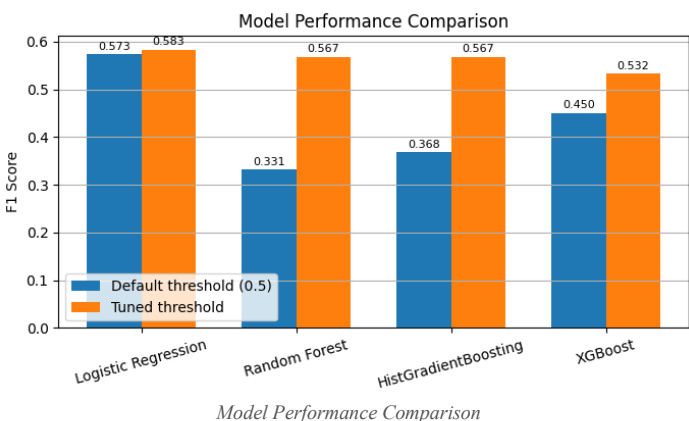
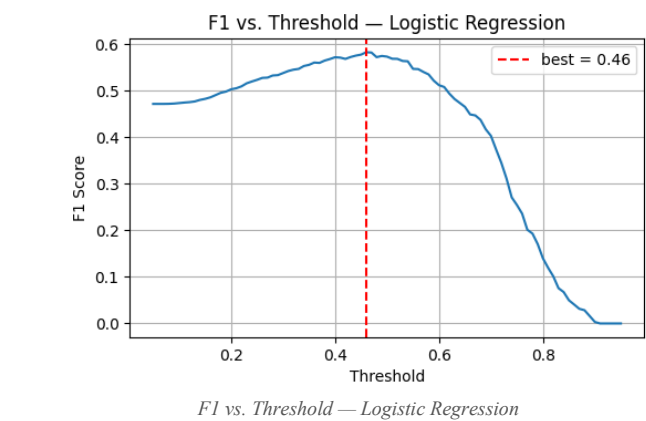
Models Evaluated

- Logistic Regression (class-weighted, default threshold 0.50)
- Random Forest Classifier (class-weighted)
- HistGradientBoosting Classifier
- XGBoost Classifier (scale_pos_weight for class imbalance)

All four models were compared with 5-fold stratified cross-validation, then refit on a held-out validation split (80/20) to search for each model's optimal decision threshold (0.05–0.95 grid).

Validation Results (F1 Score)

Model	Default F1 (0.50)	Tuned F1	Best Threshold
Logistic Regression	0.5731	0.5825	0.46
Random Forest	0.3315	0.5674	0.24
HistGradientBoosting	0.3680	0.5671	0.27
XGBoost	0.4502	0.5324	0.10



Conclusion

Threshold tuning delivered the single largest performance gain across every model — tree-based models in particular were unusable at the default 0.50 cutoff (F1 as low as 0.33) but became competitive once tuned. Logistic Regression achieved the best overall validation score (F1 = 0.5825) at an optimized decision threshold of 0.46, and was selected as the final model. It was refit on the full training set and used to generate predictions on the private test set at this threshold.